Jihyung Kim

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Research areas: Quantum query complexity and Fourier analysis of Boolean functions, in particular the Fourier growth of quantum algorithms with few clean qubits (DQC_k) or few rounds of adaptivity.

Teaching areas: Deep Learning, Data Science, Data Structures and Algorithms, and AI literacy education.

Academic Employment

2026 – Lecturer, **Sahmyook University**, Department of Artificial Intelligence Convergence, Seoul, South Korea.

Education

2025 – 2026 **M.C.S.** in **Computer Science**, **University of Illinois Urbana-Champaign**.

2021 – 2023 **Ph.D. program** in **Computer Science**, **University of Central Florida** (withdrew).

2014 – 2021 **B.S.** in **Information Science**, **University of Arizona**.
(Minor in *Mathematics*)

Research Experience

2026 – Independent research in **Quantum Query Complexity**.
Fourier growth of DQC_k algorithms (in progress). Studying level-two Fourier bounds for quantum query algorithms with only k clean qubits and the query lower bounds they give, building on Girish, "Fourier Spectrum of Noisy Quantum Algorithms" (STOC 2026).
Adaptivity in quantum query algorithms (in progress). Studying the Fourier growth of algorithms with few adaptive rounds and the trade-off between rounds and parallel queries from Girish, Sinha, Tal, and Wu, "The Power of Adaptivity in Quantum Query Algorithms" (STOC 2024).

Graduate Research Assistant, **Socio-Technical Interaction Research (STIR) Lab**, University of Central Florida.
Contributed to a UX bootcamp project with teens (ages 13–17), co-designing effective online safety interventions while teaching UX design methods and industry-standard tools.

Undergraduate Research Assistant, **Future Reality Lab**, University of Arizona.
Researched an animated digital sketching interface for visualizing ideas and concepts in live presentation and conversation contexts.
Participated in the Sciences Designs Vertically Integrated Project (VIP) — advancing the UA Holodeck and developing health- and well-being-centered interactive experiences for everyday environments.

Industry Experience

Aug 2023 – Jan 2026 **Stealien Inc.** (cybersecurity firm), Senior Strategic Planner, Management Support Office, Seoul.
Led corporate strategy planning and business-environment improvement initiatives.
Identified and maintained external partnerships and collaborative relationships.
Planned new business initiatives and formulated new-market entry strategies.

Teaching

"AI Literacy and Problem Solving"
Course instructor at **Sahmyook University** (3 credits), Spring 2026.

"Data Structures and Algorithms"
Course instructor at **Sahmyook University** (3 credits), Spring 2026.

"Data Structures"
Course instructor at **Sahmyook University** (English-medium, 3 credits), Fall 2026.

"Introduction to Data Science"
Course instructor at **Sahmyook University** (English-medium, 3 credits), Fall 2026.

"Applied Deep Learning"
Course instructor at **Sahmyook University** (3 credits), Fall 2026.

"Foundations of HCI (CAP 3104)"
Graduate Teaching Assistant at **University of Central Florida**, Fall 2021 & Summer 2022.
50–60 undergraduates (Instructor: Mike McAlpin). Human-centered design principles, interaction modalities, design guidelines and heuristics, and interface evaluation techniques.

"Security in Computing (CIS 3360)"

Graduate Teaching Assistant at **University of Central Florida**, Summer 2022.
247 undergraduates (Instructor: Mike McAlpin). Applied security theory (legal and human factors), malware, intrusion patterns and tools, Windows/Unix/TCP-IP and application vulnerabilities, and detection, policy, and assurance.

"Program Analysis (COP 5021)"

Graduate Teaching Assistant at **University of Central Florida**, Spring 2022.
17 graduate students (Instructor: Ali Almalki). Static program analysis theory — data-flow analysis, abstract interpretation, and type-and-effect systems — with program analysis automation tools.

"Computer Understanding of Natural Language (CAP 6640)"

Graduate Teaching Assistant at **University of Central Florida**, Spring 2022.
51 graduate students (Instructor: Liqiang Wang). Advanced natural language processing course focused on developments of the preceding eight years.

"Introduction to Programming Language"

Volunteer Instructor at **Pima Community College** (Adult Basic Education for College and Career), Jan – Jun 2021.
Programming fundamentals — input/output, logic, decision structures, and loops — applying design thinking in open-ended projects with strategies such as Think-Pair-Share and pair programming.

"Introduction to Computer Science"

Classroom Volunteer / Teaching Assistant at **Ha:san Preparatory & Leadership School**, Aug – Dec 2017.
Taught Snap! programming to high school students; supported building a sustainable CS program through team-teaching that connected CS industry professionals with classroom teachers.

Technical Skills

Machine Learning: PyTorch, Keras, TensorFlow

Data Science: NumPy, pandas, scikit-learn, matplotlib, spaCy

Web / Mobile: Flask, React, Android Studio

Developer Tools: Git, UNIX command line

Military Service

Mar 2015 – Dec 2016 **Republic of Korea Army**, Sergeant (honorably discharged).